```
--- Running experiment with 0 layers frozen ---
Some weights of BigBirdForSequenceClassification were not initialized from the model checkpoint at google/bigbird-roberta-base and are newly initialized: ['classifier.dense.bias', 'classifier.dense.weight', '
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
/tmp/ipython-input-4194314837.py:47: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trainer.__init__`. Use `processing_class` instead.
  trainer = Trainer(
```

[3201/3201 1:49:59, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.397100	0.358049	0.879925	0.879593
2	0.216900	0.444847	0.895872	0.895868
3	0.138400	0.509706	0.895872	0.895861

[134/134 01:25]

```
--- Running experiment with 6 layers frozen ---
Some weights of BigBirdForSequenceClassification were not initialized from the model checkpoint at google/bigbird-roberta-base and are newly initialized: ['classifier.dense.bias', 'classifier.dense.weight', '
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
/tmp/ipython-input-4194314837.py:47: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trainer.__init__`. Use `processing_class` instead.
  trainer = Trainer(
Frozen encoder layer 0
Frozen encoder layer 1
Frozen encoder layer 2
Frozen encoder layer 3
Frozen encoder layer 4
Frozen encoder layer 5
```

[3201/3201 1:41:01, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.365800	0.309121	0.888368	0.887951
2	0.188800	0.385721	0.901501	0.901500
3	0.225300	0.369256	0.902439	0.902417

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```
--- Running experiment with 12 layers frozen ---
Some weights of BigBirdForSequenceClassification were not initialized from the model checkpoint at google/bigbird-roberta-base and are newly initialized: ['classifier.dense.bias', 'classifier.dense.weight', '
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
/tmp/ipython-input-4194314837.py:47: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trainer.__init__`. Use `processing_class` instead.
  trainer = Trainer(
Frozen encoder layer 0
Frozen encoder layer 1
Frozen encoder layer 2
Frozen encoder layer 3
Frozen encoder layer 4
Frozen encoder layer 5
Frozen encoder layer 6
Frozen encoder layer 7
Frozen encoder layer 8
Frozen encoder layer 9
Frozen encoder layer 10
Frozen encoder layer 11
```

[3201/3201 1:32:02, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.640200	0.642248	0.630394	0.615765
2	0.598800	0.594413	0.723265	0.722232
3	0.597500	0.580796	0.722326	0.720982

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```
--- All experiments completed. Results: ---
Frozen Layers: 0, Metrics: {'eval_loss': 0.5501995086669922, 'eval_accuracy': 0.8761726078799249, 'eval_f1': 0.8761617100371747, 'eval_runtime': 86.5507, 'eval_samples_per_second': 12.316, 'eval_steps_per_sec
Frozen Layers: 6, Metrics: {'eval_loss': 0.47887203097343445, 'eval_accuracy': 0.8846153846153846, 'eval_f1': 0.8846152830761514, 'eval_runtime': 86.4719, 'eval_samples_per_second': 12.328, 'eval_steps_per_se
Frozen Layers: 12, Metrics: {'eval_loss': 0.6108394861221313, 'eval_accuracy': 0.6829268292682927, 'eval_f1': 0.6813069398008101, 'eval_runtime': 86.365, 'eval_samples_per_second': 12.343, 'eval_steps_per_sec
```